Controlled Chaotic Neural Networks for Pattern Retrieval

An Implementation-Oriented Interpretation

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Disclaimer

This document provides an independent implementation and interpretation of a class of chaotic neural network models for pattern retrieval. It does not reproduce or claim authorship of any published work. The focus is on understanding, reformulating, and implementing the underlying mathematical structure.

1 Introduction

Chaotic neural networks (CNNs) constitute a class of nonlinear dynamical systems capable of exhibiting rich temporal behaviors such as convergence, oscillation, and chaos. These properties make them attractive for associative memory applications, where stored patterns must be retrieved from incomplete or noisy inputs.

However, the same nonlinear dynamics that enable exploration may also prevent reliable convergence. Therefore, controlling the balance between chaotic behavior and stability is essential.

This document presents a mathematical formulation and interpretation of such systems, with emphasis on their dynamical structure and control mechanisms.

2 Chaotic Neural Network Model

We consider a discrete-time neural network composed of M fully interconnected neurons. Each neuron is described by two internal states:

- a **feedback state** $\eta(t)$,
- a **refractoriness state** $\zeta(t)$.

The evolution of the network is given by:

$$\eta(t+1) = k_f \eta(t) + Wx(t) \tag{1}$$

$$\zeta(t+1) = k_r \zeta(t) - \alpha x(t) + a \tag{2}$$

$$x(t+1) = S\big(\eta(t+1) + \zeta(t+1)\big) \tag{3}$$

where $x(t) \in \mathbb{R}^M$ is the output vector.

The activation function is chosen as a sigmoidal mapping:

$$S(y) = \frac{1}{1 + \exp(-y/\varepsilon)} \tag{4}$$

2.1 Interpretation of the Dynamics

The feedback state $\eta(t)$ aggregates the influence of other neurons through the synaptic matrix W. It drives the system toward stored patterns and acts as a cooperative mechanism.

In contrast, the refractoriness state $\zeta(t)$ introduces inertia and resistance to rapid changes. Its magnitude plays a crucial role:

- small $\zeta(t)$ promotes convergence,
- large $\zeta(t)$ may induce oscillatory or chaotic behavior.

The interaction between these two states is the primary source of complexity in the model.

2.2 Pattern Storage Mechanism

A set of patterns $\{x^{(i)}\}_{i=1}^{N}$ is embedded into the network through the weight matrix:

$$W = \frac{1}{N}\sum_{i=1}^{N}(x^{(i)} - o)(x^{(i)} - o)^T - I \tag{5}$$

where o is the vector of ones.

This construction ensures that the stored patterns act as attractors of the network dynamics.

3 Performance Indicators

To analyze the behavior of the network, several indicators are considered.

3.1 Largest Lyapunov Exponent

The largest Lyapunov exponent quantifies sensitivity to initial conditions. A positive value indicates chaotic behavior, while a negative value corresponds to stable convergence.

Since exact computation is not feasible for nonlinear systems, numerical estimation methods based on perturbation are typically used.

3.2 Quasi-Energy Function

An energy-like quantity is defined as:

$$QE(t) = -\frac{1}{2}x^T(t)(W - W \circ I)x(t) - a^T x(t) \tag{6}$$

Although not a true physical energy, this function provides insight into long-term behavior. Periodic or oscillatory dynamics are reflected in its evolution over time.

3.3 Retrieval Performance Metrics

The associative capability of the network can be evaluated using:

- the period of the output sequence,
- the number of retrieved stored patterns (RFBP),
- the number of retrieved reversed patterns (RFRBP).

These metrics depend on the initial condition and reflect the effectiveness of pattern retrieval.

4 Control of Chaotic Dynamics

The key observation is that the refractoriness state $\zeta(t)$ governs the transition between convergence and chaos.

4.1 Control Insight

- small $\zeta(t) \Rightarrow$ convergence to stored patterns,
- large $\zeta(t) \Rightarrow$ escape from equilibrium and chaotic motion.

Therefore, controlling $\zeta(t)$ is central to regulating the network.

4.2 Control Signal

A scalar signal is defined as:

$$u(t) = \frac{1}{M} \sum_{i=1}^{M} |\eta_i(t)| \tag{7}$$

This quantity reflects how close the network is to stored patterns. Larger values indicate higher similarity.

4.3 Limitations of Classical Approaches

Classical methods define an upper bound $\bar{\zeta}$ as a function of $u(t)$, often using exponential mappings. However, such approaches are highly sensitive to parameter tuning and lack robustness.

4.4 Modified Controlled Dynamics

To improve robustness, the dynamics can be modified as:

$$\eta(t+1) = k_f \eta(t) + W x(t) \tag{8}$$

$$\zeta(t+1) = R_{\bar{\zeta}(t)} \left(k_r \zeta(t) - \alpha \beta^k \, \zeta(t) \circ x(t) + a \right) \tag{9}$$

$$x(t+1) = S\big(\eta(t+1) + \zeta(t+1)\big) \tag{10}$$

where $\circ$ denotes element-wise multiplication.

The saturation operator $R_a(\cdot)$ is defined as:

$$R_a(b) = \begin{cases} b, & |b| < a \\ a, & b \geq a \\ -a, & b \leq -a \end{cases} \tag{11}$$

4.5 Interpretation of the Modified Model

The additional damping term introduces a state-dependent attenuation mechanism. When $\zeta(t)$ grows large, this term increases and counteracts further growth.

Combined with saturation, this ensures boundedness of $\zeta(t)$.

As a result:

- excessive chaotic excursions are suppressed,
- sensitivity to parameter tuning is reduced,
- convergence behavior is improved.

5 Discussion

The modification replaces parameter-dependent control strategies with intrinsic stabilization. Instead of explicitly tuning control laws, the system dynamics enforce bounded behavior.

This leads to a more robust and general framework for controlled chaotic neural networks.

6 Conclusion

This document presented an implementation-oriented interpretation of chaotic neural networks for pattern retrieval. By analyzing the role of internal states and introducing a damping-based modification, the model achieves improved stability while preserving its expressive dynamics.

7 Illustrative Retrieval Results

In this section, the behavior of the controlled chaotic neural network is demonstrated through a set of image-based retrieval experiments. The aim is to illustrate how the proposed dynamics process noisy inputs and guide them toward meaningful stored patterns.

7.1 Model and Simulation Parameters

The parameters used in the CNN model and the retrieval experiments are summarized in Table 1. These values are selected to ensure stable yet dynamically rich behavior of the controlled chaotic neural network.

Parameter	Value	Description
N	4	Number of stored patterns
M	400	Number of neurons (dimension of state vector)
a	$\mathbf{1}$	External input vector
ϵ	0.02	Sigmoid steepness parameter
k_f	0.2	Feedback decay parameter
k_r	0.95	Refractoriness decay parameter
α	10	Refractory scaling parameter (denoted as a_p in code)
C_0	5	Control parameter (offset)
C_1	-6	Control parameter (exponential bias)
C_2	1	Control parameter (input scaling)
β	0.97	Damping coefficient in modified dynamics
k	0.05	Exponential damping exponent
T	20	Number of simulation steps
Noise level	10%	Fraction of perturbed pixels in initialization

Table 1: Parameters used in the controlled chaotic neural network simulations.

7.2 Stored Patterns

A set of $N = 4$ binary image patterns is considered. Each pattern is represented as a vector in $\mathbb{R}^M$, where M denotes the number of neurons in the network. These patterns are embedded into the network through the synaptic weight matrix using a correlation-based learning rule.

Fig. 1 shows the four stored patterns used in the experiments. These patterns define the target memory states of the network.

7.3 Experimental Setup

To evaluate the retrieval capability of the network, each experiment is initialized with a noisy version of one of the stored patterns. The noise is introduced by randomly perturbing a fraction of the pixels in the original pattern. This simulates

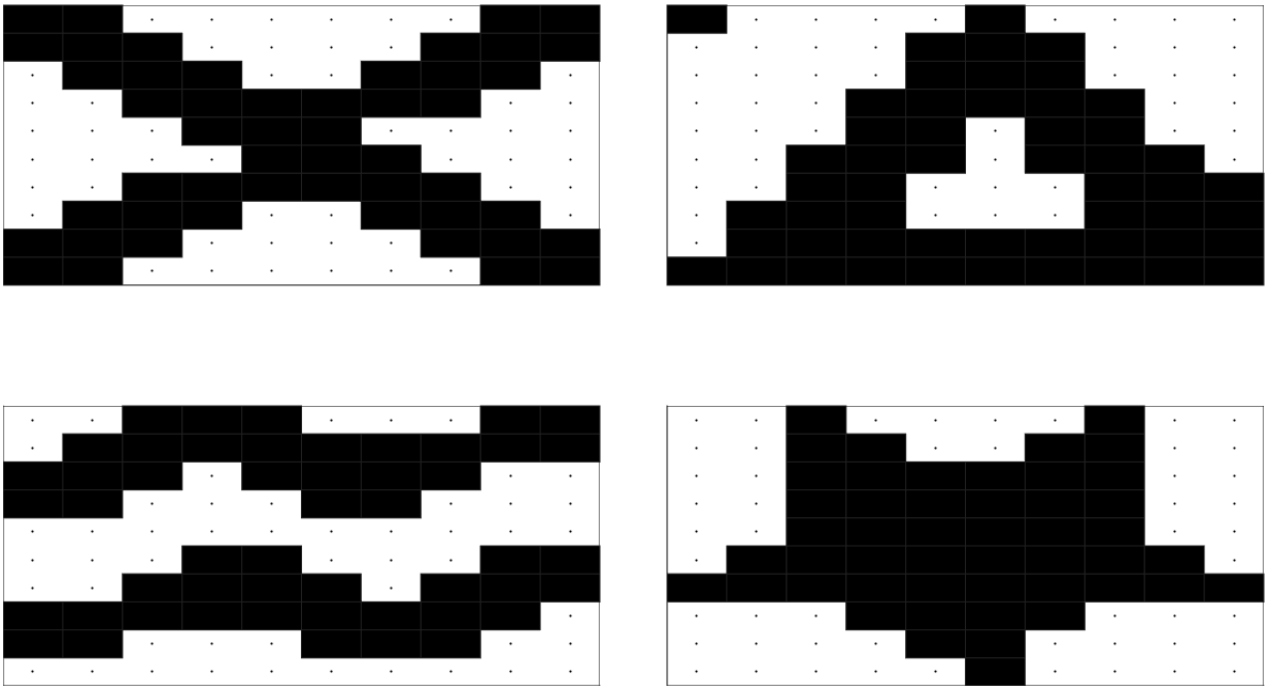

Figure 1: Stored binary patterns used in the retrieval experiments.

imperfect or corrupted inputs.

The network is then evolved according to the controlled chaotic dynamics described earlier. At each time step, the network state is updated using both the feedback mechanism and the bounded refractoriness dynamics. The evolution is observed over a finite time horizon.

7.4 State Evolution and Retrieval Behavior

Figures 2–5 illustrate the temporal evolution of the network for four different initializations. In each figure, the state at $t = 0$ corresponds to the noisy input, and subsequent panels show the network output at successive time steps.

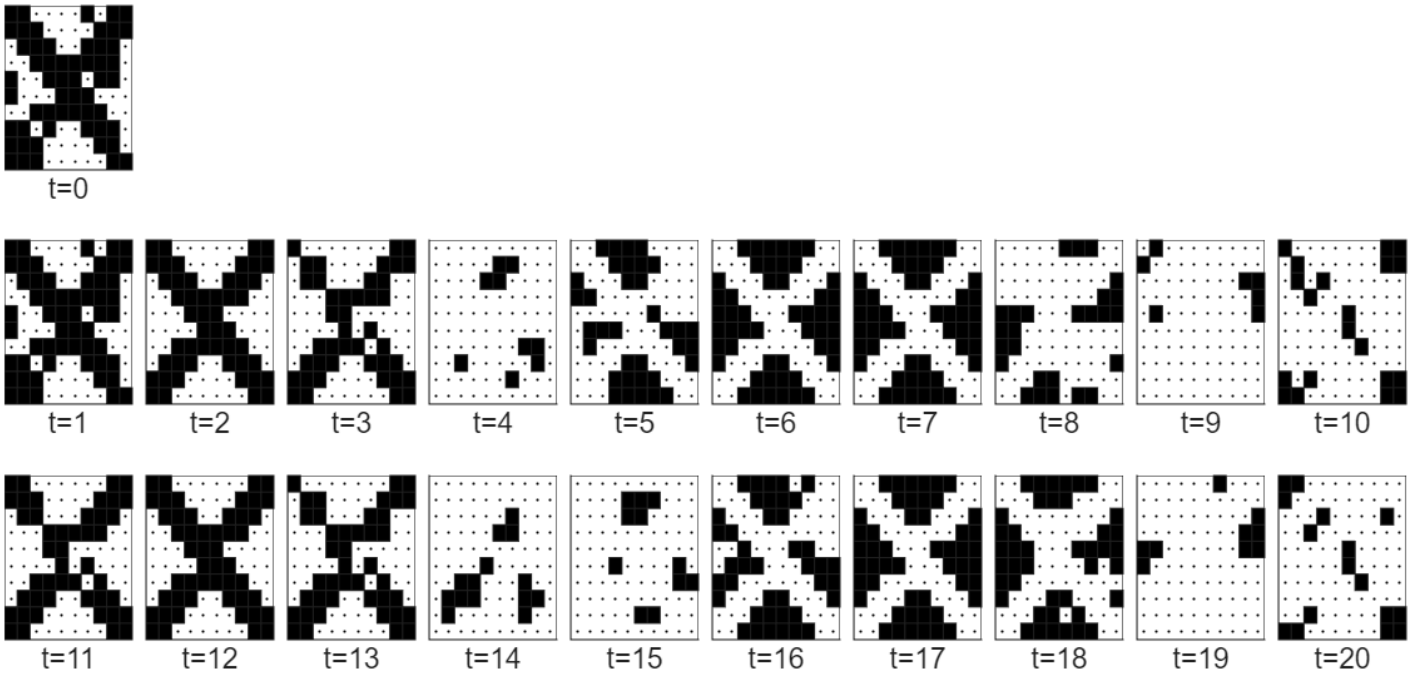

Figure 2: State evolution of the controlled CNN for a noisy initialization of pattern 1.

The results reveal several important characteristics of the proposed model.

First, the network exhibits a clear tendency to transform the noisy initial conditions into more structured patterns. Even when the initial state is significantly corrupted, the dynamics progressively enhance coherent regions in the image.

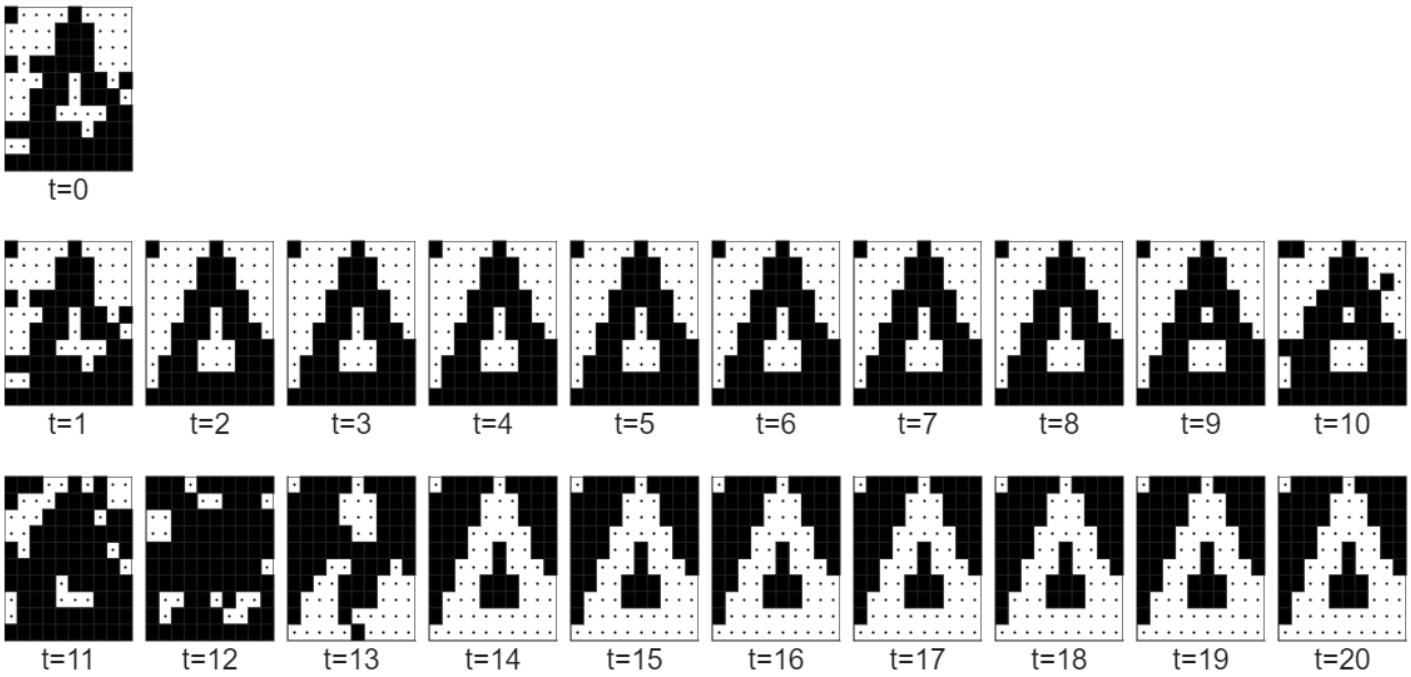

Figure 3: State evolution of the controlled CNN for a noisy initialization of pattern 2.

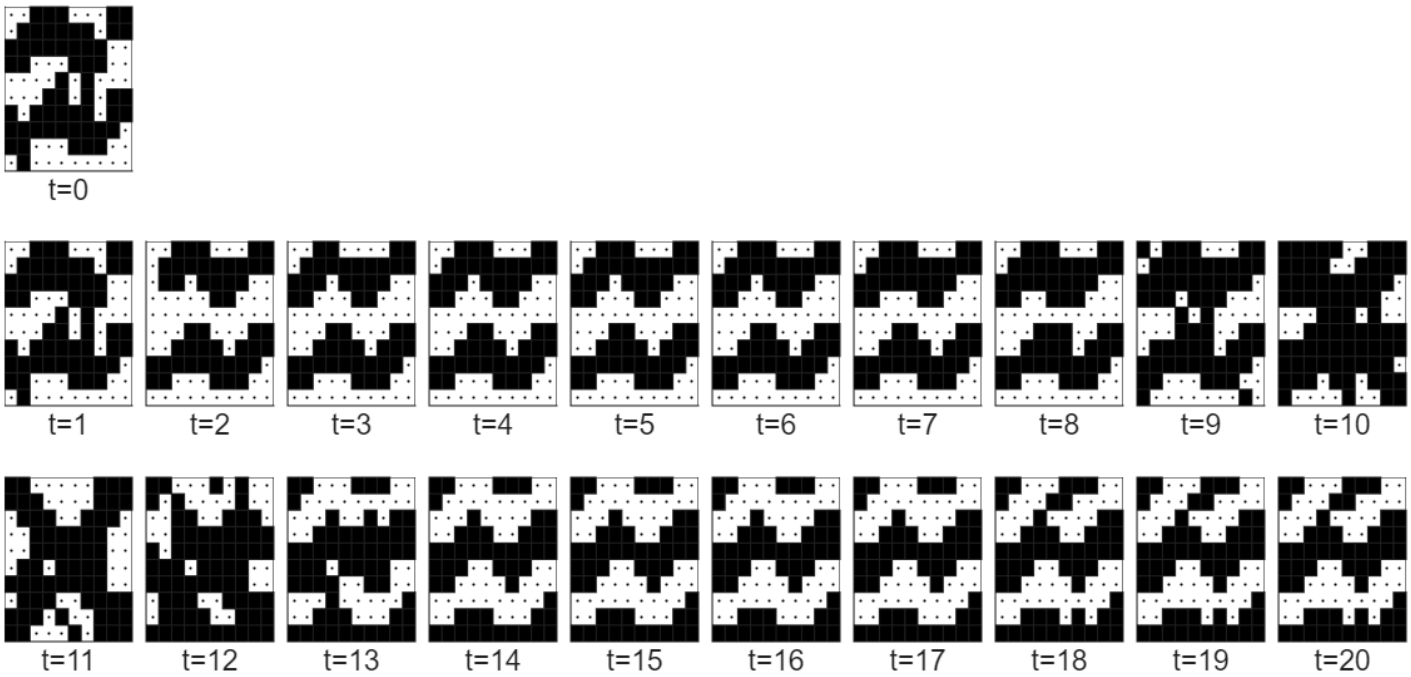

Figure 4: State evolution of the controlled CNN for a noisy initialization of pattern 3.

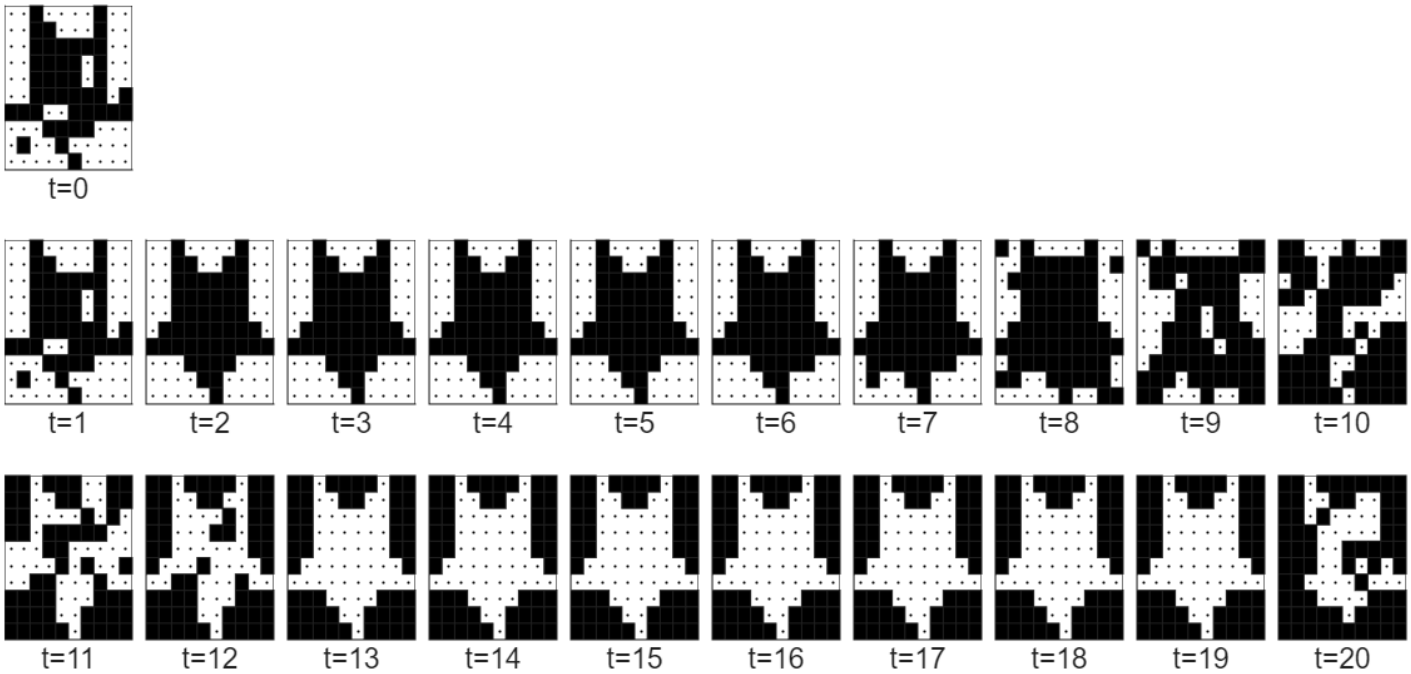

Figure 5: State evolution of the controlled CNN for a noisy initialization of pattern 4.

Second, the evolution is not purely monotonic. In some cases, the state undergoes intermediate transitions that involve partial pattern switching or reorganization. This behavior reflects the intrinsic nonlinear and controlled chaotic nature of the model.

Third, the controlled refractoriness mechanism prevents unbounded growth of internal states and stabilizes the dynamics. As a result, the network avoids divergence and maintains bounded trajectories, even in the presence of strong nonlinear interactions.

Finally, the experiments demonstrate that the model is capable of associative retrieval: the network tends to move toward patterns that resemble the stored memories. While the trajectories may involve transient oscillations or switching phases, the overall behavior supports the effectiveness of the proposed control strategy.

7.5 Discussion

The presented results highlight the balance achieved by the controlled CNN between stability and dynamic richness. Unlike purely gradient-based models, the system does not necessarily converge directly to a fixed point. Instead, it exhibits structured transient dynamics that can enhance exploration of the state space before settling into a meaningful pattern.

This property can be advantageous in applications where robustness to noise and adaptability are required. The ability to recover structured patterns from corrupted inputs, while maintaining bounded and controlled dynamics, demonstrates the practical potential of the proposed framework.